



Measuring the electron mass-charge ratio with trajectories in a uniform magnetic field

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The charge-mass ratio of the electron was one of the first measured characteristics of the electron. The original experiment that determined this ratio is replicated by measuring the radius of the electron trajectory in a uniform magnetic field. By relating this to the voltage of the electron gun and current of the coils producing the magnetic field, the mass-charge ratio is found to be $4.8 \pm 0.1 \times 10^{-12} \text{ kg/C}$.

I. INTRODUCTION

The discovery of the electron resulted from examining electrical discharges in gas. Subsequent experimentation led to the derivation of the electron charge-mass ratio, the first quantified characteristic of the particle. These measurements paved the way for future research on electrons and other particles.

Experiments by Arthur Schuster examined electrical discharge in a cathode tube and the subsequent glow. He was credited with describing the trajectory of this glow and in deriving the charge-mass ratio of the electrical carrier particles within the glow, earning him the title of “almost discovering the electron” [1]. However, J.J. Thomson, who was working on similar experiments, criticized Schuster for not making any quantitative measurements. Thomson’s experiments paved the way for his discovery of the electron. In 1897, he made the first measurements of the electron charge-mass ratio [2].

An experiment is conducted to replicate the results of J.J. Thomson in measuring the charge-mass ratio of an electron. This method involves using an electron gun in a uniform magnetic field to measure the radius of the trajectory of the emitted electron beam. The radius, voltage of the electron gun, and current in the magnetic coils can be used to calculate this ratio.

II. INTERPRETING THE ELECTRON TRAJECTORY RADIUS

To measure the charge-mass ratio of the electron, an electron gun is inserted between a pair of Helmholtz coils, identical circular loops of wire producing a uniform magnetic field [3]. The electron gun releases electrons perpendicular to the magnetic field produced by the Helmholtz coils. Since an electric field is not present, a bulb at the center of the experiment contains the emitted electrons as they travel in a circular orbit. A ruler at the back of the experiment allows for measuring the radius of the trajectory of the electrons. The voltage of the electron

gun and the current through the coils are also measured.

In performing this experiment, data was gathered by changing the coil current and measuring the radius of the trajectory at a certain voltage of the electron gun. Then, at a later trial, the voltage of the electron gun is switched to a different value, and the whole process repeated. Consequently, data was acquired on how the voltage of the electron gun and coil current affect the radius of the trajectory.

An electron with velocity v perpendicular to a uniform magnetic field B experiences the Lorentz Force. In the absence of an electric field, this force F is described as:

$$F = qvB, \quad (1)$$

where q is the charge of the electron. Since no electric field is present, the electron trajectory is a circle described by centripetal force. Since the electron is in a circular orbit, eq. (1) is set equal to the centripetal force:

$$\frac{mv^2}{r} = qvB, \quad (2)$$

with the electron’s mass m and orbit radius r .

The magnitude of the magnetic field is calculated as

$$B = I(7.81 \times 10^{-4} \text{T/A}), \quad (3)$$

where I is the coil current. The electron charge-mass ratio needs to be solved in terms of the radius of its trajectory r , the voltage of the electron gun V , and the magnetic field magnitude B . The electron gun emits the electrons with a kinetic energy K equal to that gained by the electrons while in the gun:

$$K = \frac{1}{2}mv^2 = qV. \quad (4)$$

By combining and rearranging eqs. (3) and (4), the electron charge-mass ratio is found:

$$\frac{q}{m} = \frac{2V}{r^2 B^2}. \quad (5)$$

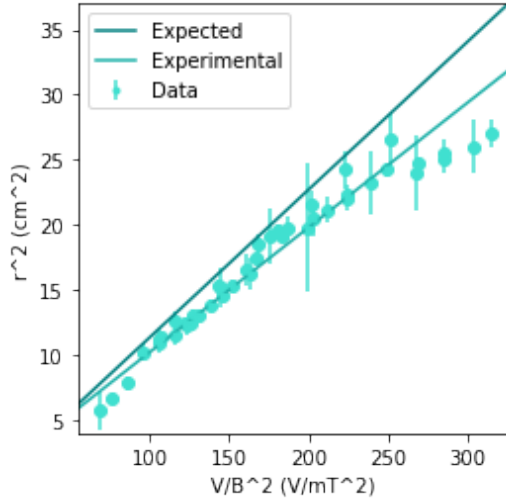


FIG. 1. The relationship described in eq. (6). The slope is weighted by uncertainty and is the mass-charge ratio of an electron, labeled as experimental data. The accepted values for the mass and charge of an electron are plotted as the expected slope.

Rearranging eq. (5) gives

$$r^2 = \left(2 \frac{m}{q}\right) \left(\frac{V}{B^2}\right), \quad (6)$$

a linear relation when plotted. This form is important, as the slope is twice the inverse of the electron charge-mass ratio, giving a direct way to determine this value. The relationship in eq. (6) is shown in fig. 1.

In fig. 1, the experimental data, a line of best fit weighted by uncertainty, and the scientifically accepted charge-mass relationship are compared. The uncertainty in the data generally grows as the x-axis values increase. This is because the uncertainty is related by a square root, so as the x-axis increases, so does the uncertainty.

Since, according to eq. (6), the slope of the plot is twice the inverse of the charge-mass ratio, this ratio can be calculated. Since calculating this ratio from the slope would give a large value of uncertainty, the inverse of the ratio, the mass-charge ratio, is calculated by dividing the slope by two to get $4.8 \pm 0.1 \times 10^{-12}$ kg/C. Comparatively, the scientifically accepted value of the electron mass-charge ratio is 5.69×10^{-12} kg/C. The actual value is not within the uncertainty range of the experimental value. However, these values are still very close, and

more measurements with higher accuracy would be able to improve these results.

III. CONCLUSION

The experiment originally conducted by J.J. Thomson to measure the charge-mass ratio of the electron was replicated, giving a value of $4.8 \pm 0.1 \times 10^{-12}$ kg/C. This was achieved by placing an electron gun within a pair of Helmholtz coils and measuring the radius of the produced electron trajectory.

Future work involves improving the experimental results. The method used for calculating the electron trajectory radius resulted in high values of uncertainty, so conducting more trials or using a different technique would result in more accurate results. The result originally attained by Thomson, however, was significant in paving the path to learning key properties of the electron. This work helped him prove the status of electrons as particles inside the cathode tubes instead of the prevailing theory of negatively charged atoms.

Appendix A: Uncertainty analysis

The uncertainty is derived from the measurement of the electron trajectory radius. This was completed by lining one's eye up with the reflection of the eye in a mirror behind the apparatus and with the edge of the trajectory circle. The measurements were added to find the circle's diameter and were completed by each lab partner. Then, these measurements were averaged together and divided by two to get the radius. The uncertainty was determined as the difference between the diameter measurements taken by each lab partner. If no difference was found, the average uncertainty among all other trials was recorded instead. Then, the uncertainty in diameter was divided by two to get the radial uncertainty. Since the uncertainty σ_r was in the radius r , the formula:

$$\delta_r = 2r\sigma_r \quad (A1)$$

gives the uncertainty in the radius squared δ_r . This is the uncertainty used in fig. 1.

Variable	Value	Units
Experimental Slope	0.096 ± 0.003	$\text{cm}^2\text{mT}^2\text{V}^{-1}$
Expected Slope	0.114	$\text{cm}^2\text{mT}^2\text{V}^{-1}$
R^2	0.951	

TABLE I. Results of fitting

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